

BufferedReader and FileReader

👉 **BufferedReader**

BufferedReader is a Java class that provides efficient reading of text from character input streams. It uses an internal buffer to read characters, making I/O operations more efficient compared to reading one character at a time. This is especially beneficial when reading large files.

- Improved performance through buffering
- Convenient methods like **readLine()** to read entire lines of text
- More efficient than reading character by character with **FileReader** alone

Example:

```
import java.io.*;

public class LaunchIO {
    public static void main(String[] args) throws IOException {

        String filePath = "D:\\telusko\\InputOutput\\java.txt"; // Path to the file we want to read

        FileReader fileReader = null; // Declares FileReader reference
        BufferedReader bufferedReader = null; // Declares BufferedReader reference

        try {
            // Create a File object representing the physical file
            File file1 = new File(filePath);

            // Create a FileReader to read the file character by character
            fileReader = new FileReader(file1);

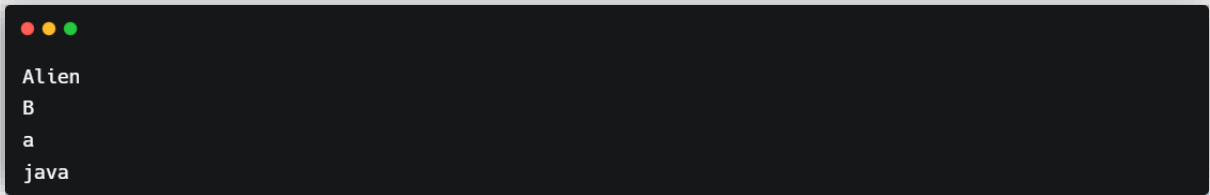
            // Wrap FileReader with BufferedReader for more efficient reading
            bufferedReader = new BufferedReader(fileReader);

            // Read the first line from the file
            String str = bufferedReader.readLine();

            // Continue reading until we reach end of file (null)
            while (str != null) {
                System.out.println(str); // Print the current line
                str = bufferedReader.readLine(); // Read the next line
            }

        } catch (Exception e) {
            System.out.println("Some problem"); // Simple error handling
        } finally {
            // Close the resources to prevent memory leaks
            bufferedReader.close();
        }
    }
}
```

Output:



```
Alien
B
a
java
```

👉 Explanation

- **Setting Up:** We specify the path to our file and declare our reader objects.
- **Creating Objects:**
 - `File` object represents the actual file on the disk
 - `FileReader` opens a connection to the file for reading characters
 - `BufferedReader` wraps around the `FileReader` to make reading more efficient
- **Reading Process:**
 - `bufferedReader.readLine()` reads an entire line of text (until it hits a line break)
 - The `while` loop continues reading lines until `readLine()` returns `null` (end of file)
 - Each line is printed to the console
- **Proper Resource Management:**
 - The `finally` block ensures we close our resources even if an error occurs
 - Closing resources is important to prevent memory leaks